"If anyone uses this for I.Q. enhancement, put the scientist in jail."[1] These strong words from Dr. He Jiankui illustrate one of the passionate stances surrounding gene editing and a modern example of the ethical discussion around human value. Although Dr. He successfully engineered three gene-edited babies, he still insists that ethical boundaries must exist regarding how gene-editing technologies are used. His argument points toward the ongoing societal struggle of determining who decides what traits are desirable or valuable in human beings. Society has debated the ethics of human value in a variety of ways through the centuries and recently through the eugenics movement in the United States and Europe. Gene editing has sometimes been compared to eugenics because both intend to eliminate supposedly undesirable characteristics, even though what counts as "undesirable" is shaped by an individual's background and perspective. While the technologies behind gene-editing and eugenics differ, both are dubious and both are profoundly intertwined with data science. Eugenics itself represents an uncomfortable era in the history of the modern world and in the development of data science. Many of the foundational tools of statistics and data analysis were developed by eugenicists.[2] This historical entanglement illustrates the necessity of careful ethical reflection by those advancing any field, whether mathematics, psychology, agriculture, or data science. Using eugenics as a historical case and tracing its influence forward into modern data science, this essay argues four primary claims. First, data tells nothing by itself; it must form the basis of a story if it is to be more than numbers. Second, every narrative is necessarily driven by a viewpoint because there is no true neutrality. Third, data-driven narratives magnify both good and harm at a population level. Finally, these three claims regarding data and data science will be related to the importance of taking action each day to analyze data in a thoughtful, ethical manner.

In the 1860s, eugenicist Francis Galton argued that statistical evidence demonstrated the inherent inferiority of certain groups of human beings.[3] In doing so, he fell prey to the fallacy that data speaks for itself. Data actually consists only of measurements and relationships and becomes meaningful only through interpretation. Galton made valuable contributions to the field of statistics, yet his conclusion that statistical analysis justified eugenics represents only one possible and certainly dubious interpretation of the data he collected.[4] In his 1869 work *Hereditary Genius*, Galton compiled genealogies of "illustrious" men and observed that many had relatives who were also socially prestigious.[5] He concluded from this correlation that prestige and high achievement were heritable traits. From a modern perspective, it is evident that this conclusion was shaped by selection bias. Galton's sample disproportionately included men who benefited from generational wealth, social power, and access to elite educational resources, all of which influence achievement unrelated to heredity. Galton further conflated correlation with causation by assuming that statistical association implied biological determination. His narrative presupposed that shared achievement among relatives was caused by genetic inheritance instead of shared environments, opportunities, and social structures. When extending his analysis, Galton measured physical characteristics and applied the

[1] Higgins
[2] Aylward
[3] Higgins
[4] Ibid.
[5] Kennedy-Shaffer

statistical normal distribution to these traits, then improperly generalized this framework to intelligence and human worth. Galtlon treated intelligence as a normally distributed, biologically fixed attribute analogous to physical traits, therefore imposing a questionable causal interpretation that was not supported by his data.[6] Galton's work illustrates how statistical tools, when embedded within an unexamined narrative, can be used to lend authority to ethically indefensible conclusions. The data itself did not require eugenics, Galton's prior assumptions shaped the story the data was made to tell.

One of the many heroes who strove to uncover the limitations of Galton's work was Horace Mann-Bond, who specifically aimed toward assisting black people rise above the narrative of black inferiority.[7] Predictably, since Galton's work established dubious causation between physical characteristics and intelligence or prestige, black people were a target of inferiority propaganda. Horace Mann-Bond shaped the data to a different narrative by bringing attention to the lack of educational resources that black people suffered in comparison to white people of prestige.[8] Instead of deciding that intelligence and prestige must be heritable, Mann-Bond utilized environmental data to find new correlations between resources and prestige. The contrast between Francis Galton and Horace Mann-Bond highlights the necessity of checking one's own narrative and assumptions before proclaiming causation. In data science, these anecdotes call attention to the societal benefits of team collaboration between individuals of varied backgrounds and viewpoints. Since data is merely numbers until it is used to shape a narrative, these data and narratives can be debated with a view toward benefitting all members of society.

This supports the next major point: that every narrative necessarily comes from a perspective. Brain researcher Tracey Tokuhama-Espinosa has found that every brain is unique, just like human faces.[9] The human brain connects new input to old and every person has different experiences from the time of conception, meaning that each human being sees the narrative that has been constructed by their experiences. Human beings are unable to be absolutely neutral when deriving a narrative around data. This means that society should tread with discernment when approaching a narrative that claims to be neutral. When considering the basis for eugenics in a 1927 textbook, Hogben beautifully asked these questions, "What is the good of the race? What is a desirable social quality? What is a 'morally and mentally fit' person?"[10] These philosophical questions underscore the impossibility of coming up with unified answers. Perhaps one segment of society believes there are clear answers to these questions and cannot understand how anyone could have different viewpoints, and yet other segments of society have the same level of certainty about polar opposite answers. Eugenics was based upon the narrative that fitness, desirable social qualities, and the good of the race was self-evident according to the data.[11] It is clear, however, that eugenics actually came from a certain perspective driving a narrative.

A further illustration of the impossibility of neutral data analysis can be found in the work of Karl Pearson. Pearson approached his statistical research with the prior assumption that

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[6] Ibid.
[7] Ibid.
[8] Ibid.
[9] Tokuhama-Espinosa
[10] Kennedy-Shaffer
[11] Aylward

Jewish individuals were intellectually inferior to Gentiles, an example of motivated reasoning operating before data analysis began. With this narrative already assumed, Pearson interpreted and structured data in ways that appeared to demonstrate the intellectual superiority of Gentile boys and girls over their Jewish counterparts.[12] Similarly to Francis Galton, Pearson made foundational contributions to statistics and mathematics. His ethical conclusions, however, show how statistical authority can be misused when analysis is guided by unexamined assumptions. Instead of allowing others' interpretation of data to challenge his beliefs, Pearson selected and interpreted evidence in ways that reinforced his preconceived narrative. History shows that Jewish populations and other population groups have been subject to terrible discrimination and abuse due to conclusions supposedly based on statistics and scientific research, which were really fore-gone conclusions supported by manipulated or missing data. Pearson's work illustrates how statistics, when separated from examination by diverse individuals, can lend legitimacy to narratives that cause profound and lasting harm.

The Jewish example leads to the third central claim of this essay, that unethical narratives supported by dubious statistical practices can magnify harm at a population scale. Eugenics, beyond its effects on any single group, provides a historically prominent illustration of how data-driven narratives can compound societal damage. In the early twentieth century, statistician Ronald A. Fisher developed analysis of variance (ANOVA), a powerful statistical technique designed to isolate and compare sources of variation within data. Fisher frequently used ANOVA within a narrative that emphasized heredity as the primary determinant of human intelligence and development.[13]

By using ANOVA to minimize or exclude environmental variation in population-level analyses, Fisher supported his narrative that intelligence is biologically fixed. Modern research, however, reveals that intelligence is shaped by the interaction of genetic, environmental, and social factors and is not immutable at birth.[14] Fisher's narrative gained influence not just because of supposed empiricism, but because of the perceived authority of statistical science, limited oversight and accountability, a lack of diverse perspectives within the field, and active government support for eugenic policies.[15] Fisher's work thus illustrates how even sophisticated statistical tools can be mobilized in support of harmful narratives when interpretation is guided by pre-conceived narratives rather than ethical scrutiny.

In comparison to the current day, the apparent authority of science and statistics still leads to the quieting of some pushback. Now, however, society members also live with the apparent authority of algorithms and varieties of artificial intelligence. The lack of understanding that most people experience regarding algorithms can lead to a dearth of accountability or oversight since even governmental authority figures can lend excessive trust to algorithms. There can be a snowball effect when societal enthusiasm for artificial intelligence leads to the demand for government support of its use in all parts of society, which can lead to particular harm in policing, educational opportunities, and financial work.

Joy Buolamwini, who founded the Algorithmic Justice League in 2016, has fought to bring the problem of magnified algorithmic harm to public attention. Her research in facial

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[12] Kennedy-Shaffer
[13] Ibid.
[14] Tokuhama-Espinosa
[15] Kennedy-Shaffer

recognition technology found that dark-skinned females are disproportionately misidentified compared to light-skinned males, highlighting issues with both gender and skin color. When government or criminal investigations place too much trust in facial recognition technology, darker-skinned people and female individuals are more likely to be mistakenly hurt in criminal prosecutions.[16] While this technology is valuable, it requires a realistic approach to know that it is imperfect and requires thoughtful oversight with future improvements. Specific improvements can be made by performing rigorous testing on algorithms across diverse populations to be sure that the technology leads to fair outcomes for all groups.[17] This disaggregation across groups is crucial because algorithms have a historical tendency to work best for the population group which creates them.[18]

Algorithms now influence wide-ranging decisions from credit approval and hiring to criminal sentencing and healthcare diagnostics. These systems affect fundamental aspects of human well-being; therefore, preventing harm requires interdisciplinary teams that incorporate diverse perspectives. Human beings are often blind to their own narratives, and without such safeguards, data can be used to reinforce harmful assumptions at scale.

A major lesson that can be learned from the eugenics-related beginnings of data science is that new developments in any field should be approached with caution. This is not to say that we shouldn't encourage forward motion; we must make progress. Progress cannot be made without a willingness to make mistakes and even fail. However, we must, as a society, be willing to include diverse viewpoints in order to prevent harm when possible. Most importantly, we must develop the essential love of truth that leads us to admit that we were wrong as individuals, as a team, or as a society. Once we can admit we were wrong, we can learn from our collective mistakes, make corrections and reparation, and move forward with greater strength. The struggle for fairness is one that is worth engaging, never giving up due to mistakes. We must not fear to walk forward carefully as data scientists.

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[16] Buolamwini
[17] Monroe-White
[18] Kennedy-Shaffer

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